

Question Paper

Exam Date & Time: 04-Mar-2025 (02:30 PM - 04:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

Modern Databases for Big Data [BDA 5202]

Marks: 50

Duration: 90 mins.

MID-SEMESTER TEST

Answer all the questions.

State any assumptions clearly and precisely.

- 1) A group messaging system allows members of a group to exchange text messages. The system does not limit the number of groups that exist in the system. However, the system is required to enforce the following constraints: (10)
(C1) A group must have a unique name in the entire system.
(C2) A group can have at most 200 members.
(C3) A user can create at most 10 groups.
(C4) A user can be a member of at most 16 groups.
(C5) A group can have one or more admins.
(C6) There is no limit on the number of admins for a group.
(C7) Each text message can be at most 512 bytes in size.
(C8) A response message may refer to a prior message.
(C9) There are two types of messages: Simple and Critical.
Create a set of syntactically correct definitions for the necessary collections and documents using MongoDB's schema specification language.
- 2) Differentiate between the following concepts: (10)
(a) Replica Set and Sharded Cluster in MongoDB architecture.
(b) Eventual consistency guarantee versus ACID guarantee.
- 3) Given the group messaging system you described in question (1) above, justify the readConcern and writeConcern you may want to use while performing the following operations: (10)
(a) making an existing member the admin of the group.
(b) reading messages from a group.
(c) sending a critical message to a group.
(d) deleting a message for the entire group.
(e) deleting a group for which you are an admin.
- 4) For the problem statement in question (1) above, define SQL tables, indices, and constraints that you may require to support a fully functional messaging system. (10)
- 5) Analyze the importance of the following principles and/or ideas: (10)
(a) CAP theorem
(b) BASE transaction model (as against ACID model)

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